

MITO, Japan

WMO: 476290

Lat: 36.3811N

Lon: 140.4681E

Elev: 31

StdP: 100.95

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.8	-12.6	1.3	4.5	-11.0	1.5	4.0	8.3	5.7	7.1	5.9	1.5	340	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	33.1	25.7	31.7	25.3	30.3	24.8	26.4	31.3	25.8	30.1	25.2	29.1	3.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.3	28.8	24.6	19.7	28.2	24.1	19.1	27.7	82.7	31.5	80.0	30.3	77.6	29.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	6.0	5.2	DB	-6.1	35.5	1.1	1.2	-6.9	36.3	-7.5	37.0	-8.1	37.6	-8.9	38.5	(4)
			WB	-6.9	27.6	1.1	0.6	-7.7	28.1	-8.3	28.5	-8.9	28.8	-9.7	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	3.5	4.2	7.4	12.4	17.2	20.5	24.8	26.1	22.5	16.9	11.0	5.9
	DBStd	8.27	2.15	2.56	3.14	3.34	2.85	2.73	3.01	2.57	3.10	2.96	3.01	2.75
	HDD10.0	630	200	164	96	14	0	0	0	0	0	24	131	
	HDD18.3	2112	458	396	341	181	54	9	0	3	63	221	385	
	CDD10.0	2246	0	1	14	86	224	316	459	500	375	214	54	4
	CDD18.3	686	0	0	0	2	20	75	201	242	128	19	0	0
	CDH23.3	5002	0	0	0	14	110	338	1563	2152	762	64	0	0
	CDH26.7	1595	0	0	0	1	13	68	544	769	191	9	0	0
Wind	WSAvg	2.3	2.1	2.3	2.6	2.7	2.5	2.3	2.3	2.3	2.3	2.1	1.8	1.9
Precipitation	PrecAvg	1374	51	52	86	116	142	146	149	130	192	175	83	51
	PrecMax	1901	122	171	150	223	337	249	319	394	412	507	205	187
	PrecMin	1111	4	8	40	28	77	63	30	20	10	12	0	0
	PrecStd	177	34	35	32	54	54	44	70	86	91	113	49	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.4	20.9	25.4	28.5	30.7	34.4	35.0	32.3	27.9	21.3	17.4
		MCWB	8.9	12.2	12.7	15.7	19.2	23.3	25.7	26.6	24.6	21.9	15.6	13.0
	2%	DB	11.7	13.3	18.1	22.5	26.1	28.6	33.0	33.5	30.6	24.9	19.5	14.8
		MCWB	6.5	8.0	12.0	15.5	18.5	22.2	25.6	26.1	24.5	19.7	14.7	10.1
	5%	DB	10.1	11.3	15.9	20.5	24.3	26.8	31.5	32.1	28.9	23.3	18.0	13.1
		MCWB	4.9	6.2	10.8	14.5	17.6	21.1	25.3	25.7	24.0	18.9	13.7	8.7
	10%	DB	8.7	9.5	13.8	18.7	22.7	25.2	29.8	30.7	27.2	21.8	16.7	11.5
		MCWB	4.0	5.0	9.5	13.4	17.2	20.8	24.7	25.2	23.2	17.7	13.0	7.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	13.3	15.1	19.6	22.0	24.6	27.1	27.6	26.1	22.9	17.7	14.7
		MCDB	13.0	16.5	17.4	22.5	25.4	29.3	32.4	33.2	29.6	25.6	19.7	16.1
	2%	WB	7.6	9.1	13.5	17.1	20.2	23.5	26.2	26.7	25.2	21.2	16.2	11.4
		MCDB	10.2	11.9	16.8	20.5	23.7	27.1	31.3	31.7	28.6	23.8	18.0	13.7
	5%	WB	5.9	7.3	11.8	15.5	19.1	22.5	25.5	26.1	24.6	19.9	15.0	9.5
		MCDB	9.1	10.2	15.1	19.0	22.6	25.5	30.1	30.6	27.9	22.2	16.9	12.1
	10%	WB	4.6	5.7	9.8	14.4	18.1	21.3	24.9	25.6	24.0	18.6	13.7	8.0
		MCDB	7.9	8.6	12.9	17.6	21.2	24.1	28.9	29.6	26.9	21.0	16.0	10.7
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.0	10.1	9.9	8.6	6.8	6.9	7.0	7.0	8.1	9.6	10.2
		MCDBR	12.2	12.5	13.6	13.2	11.8	9.7	9.5	9.4	8.6	9.2	10.9	11.6
		MCWBR	7.7	8.2	8.6	7.4	5.9	4.5	3.8	3.6	3.8	5.2	7.0	7.7
	5% WB	MCDBR	10.9	11.4	12.2	10.7	9.4	8.0	8.7	8.5	7.9	7.5	8.9	10.1
		MCWBR	7.8	8.5	9.1	7.0	5.7	4.3	3.8	3.5	3.8	5.1	6.8	7.7
Clear-Sky Solar Irradiance	taub	0.372	0.422	0.495	0.548	0.568	0.603	0.598	0.582	0.511	0.471	0.418	0.376	
	taud	2.247	2.098	1.921	1.846	1.852	1.836	1.899	1.930	2.087	2.116	2.182	2.256	
	Ebn at Noon	798	795	767	751	743	716	716	716	748	736	736	764	
	Edh at Noon	105	137	179	203	205	208	194	185	150	134	112	98	
All-Sky Solar Radiation	RadAvg	2.72	3.26	4.06	4.75	5.11	4.67	4.86	4.78	3.78	3.01	2.48	2.35	
	RadStd	0.15	0.30	0.22	0.42	0.50	0.36	0.76	0.51	0.30	0.28	0.22	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	N/A
Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page